

- 7 (a) Tungsten, a transition metal, is commonly used as a target metal for the production of X-rays. The energy levels of the K- and M-shells for tungsten are shown in Fig. 7.1 below.

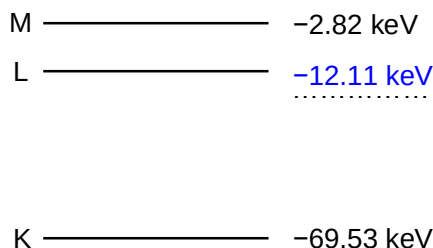


Fig. 7.1

The wavelength of the photon produced by the K_α transition is 21.65 pm .

- (i) Complete Fig. 7.1 by filling in the energy level of the L-shell for tungsten.

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{21.65 \times 10^{-12}}$$

$$= 9.187 \times 10^{-15} \text{ J} = 57.42 \text{ keV}$$

$$E_2 = -69.53 + 57.42 = -12.11 \text{ keV}$$

Examiner's comments:

A number of students rounded values prematurely, resulting in inaccurate final values.

[2]

- (ii) Determine the wavelength of the photon produced by L_α transition.

$$-2.82 - (-12.11) = 9.29 \text{ keV}$$

$$\lambda = \frac{hc}{E} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{9.29 \times 10^3 \times 1.6 \times 10^{-19}}$$

$$= 134 \text{ pm}$$

wavelength = pm [1]

The intensity of various photon wavelengths from electron bombardment of a tungsten target metal is shown in Fig. 7.2 below.

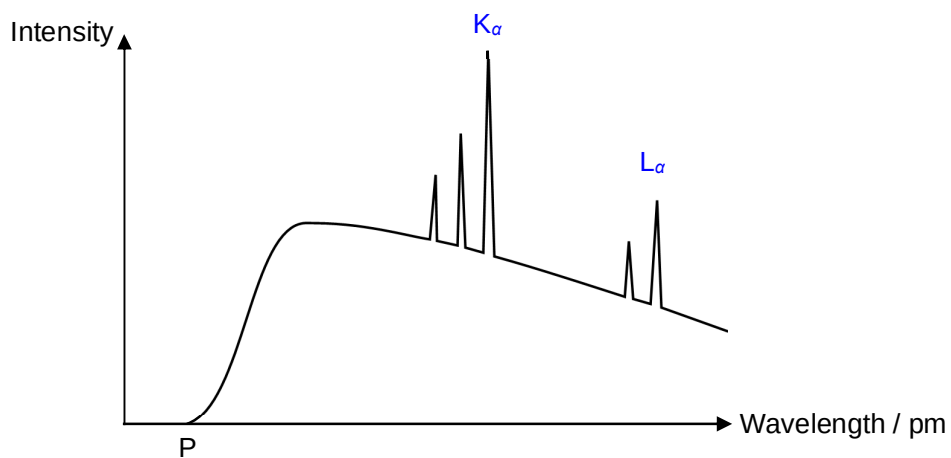


Fig. 7.2

(iii) On Fig. 7.2, label the peaks for K_{α} and L_{α} transitions. [2]

(iv) State the significance of the value of P.

indicates highest energy photons / KE of electrons [1]

no marks for "minimum wavelength" only

Examiner's comments [1]

An alarming number of students used phrasing that indicates a concept error.

Photons do not have KE, nor is KE "transferred" from electrons to photons.

- (v) With reference to Fig. 7.1, explain why the electrons bombarding the target metal must have a minimum energy of 69.53 keV to produce the spectrum shown in Fig. 7.2.

To produce the characteristic lines in the spectrum, [1]

electrons must have sufficient energy to knock out / remove K-shell electrons. [1]

The amount of energy required to bring the electron from K-shell to ionized state is 69.53 keV / the energy level of the K-shell is -69.53 eV. [1]

Examiner's comments:

Some students are still labouring under the misconception that electrons can deexcite from "infinity energy level" to ground state and produce a photon. E_{∞} indicates the electron has escaped the atom (free electron) and has a minimum energy of 0 eV [3]

- (b) Fig. 7.3 below shows a typical setup for producing such X-ray beams.

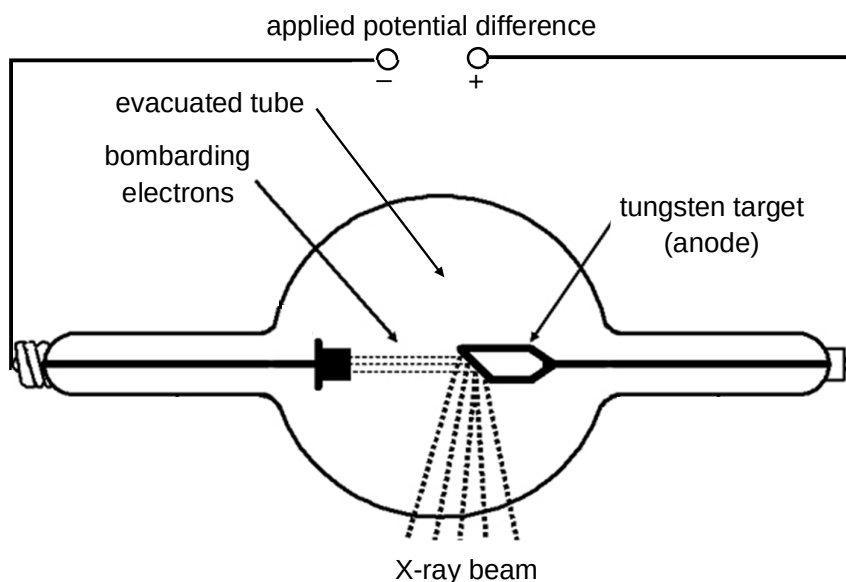


Fig. 7.3

For safety reasons, the wavelength of radiation used for medical X-rays is no shorter than 50 pm. Explain, with calculation, how the setup in Fig. 7.3 can be adjusted to ensure that the wavelength of the X-rays produced does not fall below 50 pm.

Minimum wavelength of X-rays produced is determined by KE of electrons which is, in turn, determined by the accelerating p.d. [1]

$$eV = \frac{hc}{\lambda}$$

$$= \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{50 \times 10^{-12}}$$

$$V = 2.5 \times 10^4 \text{ V} = 25 \text{ kV} \text{ [1] for working to show value of V}$$

The accelerating p.d. is to be kept under 25 kV. [1]

[3]

Section B

Answer **one** question from this section in the spaces provided.

- 8 (a) Point source P, consisting of light with wavelength 630 nm, passes through a narrow slit and is incident on a screen at a distance of 2.4 m from the slit. Fig. 8.1 below shows the variation of intensity I of the light on the screen with distance x along the screen.

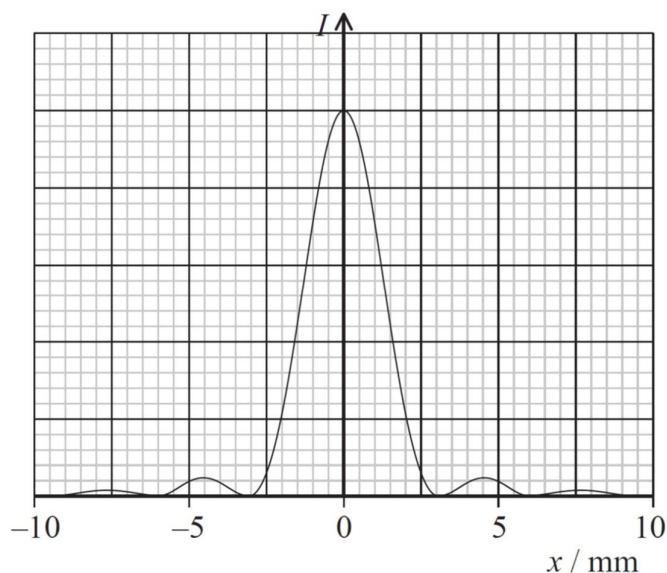


Fig. 8.1

- (i) Use Fig. 8.1 to determine the width of the slit.

$$\sin \theta = \lambda/b$$

$$\sin \theta = \frac{0.003}{2.4} = \frac{\lambda}{b}$$

$$b = 5.04 \times 10^{-4} \text{ m}$$

- (ii) State the effect on the pattern on the screen if each of the following changes is made separately:

1. the width of the single slit is reduced,

$b \downarrow$, $\sin \theta \uparrow$, $\theta \uparrow \rightarrow$ increase in width of central maximum, maxima are further lower intensity

2. the red source is replaced with another source of violet light.

$\lambda \downarrow$, $\sin \theta \downarrow$, $\theta \downarrow \rightarrow$ reduced width of central maximum, maxima are closer green and dark regions

- (b) Another point source Q emitting light of the same wavelength as P is placed 5.0 mm from P, and the two sources are arranged as shown in Fig. 8.2. The slit is now replaced with another slit of width b .

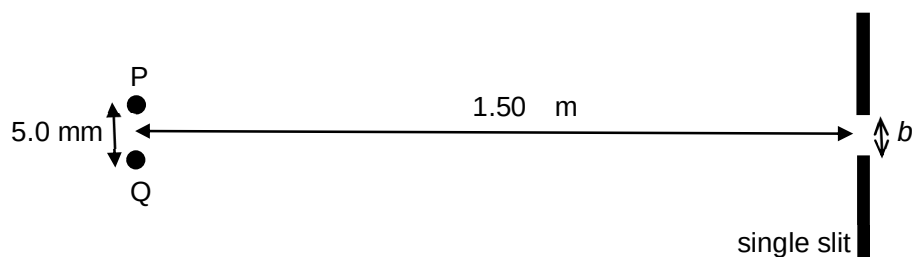


Fig. 8.2

- (i) State the condition for the image of P and the image of Q formed on the screen to be just resolved.

Rayleigh's criterion states that two images are just distinguishable if the central maximum of one diffraction pattern falls on the first minimum of the diffraction pattern of the other.

- (ii) Sketch, on Fig. 8.1, the graph for point source Q when the condition in (b)(i) is met. [1]
- (iii) Calculate the minimum width b of the slit for the two images to be just resolved. [3]

$$\tan\theta = \frac{0.0025}{1.50} = 1.6667 \times 10^{-3}$$

By small angle approximation, $\theta_{res} = 2\tan\theta$

$$\theta_{res} = \frac{\lambda}{b}$$

$$b = \frac{630 \times 10^{-9}}{2 \times 1.6667 \times 10^{-3}} = 1.89 \times 10^{-4} \text{ m}$$

$$b = \dots\dots\dots \text{ mm [2]}$$

- (c) Two microwave transmitters, A and B, are separated by a distance of 0.80 m. They each transmit a microwave of frequency 3.00 GHz.

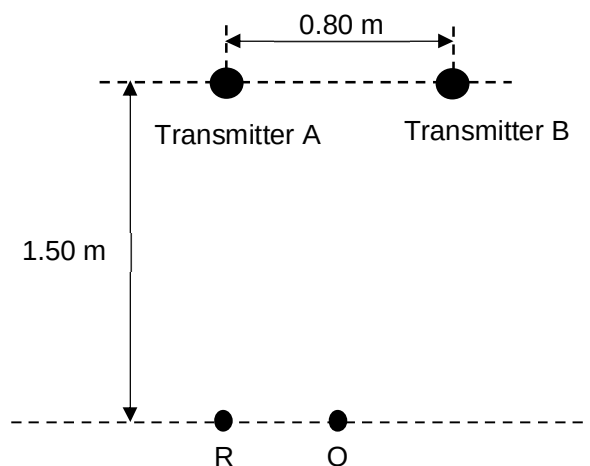


Fig. 8.3

A microwave receiver is moved along a line parallel to, and 1.50 m from, the line joining the two transmitters, as shown in Fig. 8.3. Point O lies on the line where it is equidistant from both transmitters, while point R is at a distance 1.50 m from transmitter A. As the receiver moves from O to R, a minimum signal is detected at R.

- (i) Determine the path difference between the two waves arriving at R.

$$S_2R = (0.80^2 + 1.50^2)^{1/2} = 1.70 \text{ m}$$

$$\text{Path difference} = 1.70 - 1.50 = 0.20 \text{ m}$$

- (ii) Show that the two waves are emitted in antiphase at the transmitters. [4]

$$\text{Wavelength of microwave} = c / f = (3.00 \times 10^8) / (3.00 \times 10^9) = 0.100 \text{ m}$$

$$\text{Path difference} = 0.20 \text{ m} = 2 \times \text{wavelengths}$$

$$\text{Phase difference due to path difference} = (2\lambda / \lambda) \times 2\pi = 4\pi$$

$$\text{Total phase difference} = 3\pi \text{ or } 5\pi$$

Since the signal received at R is a minimum, destructive interference must have occurred where the waves met in antiphase. Therefore, a phase difference of π must have been introduced at the sources.

$$\text{i.e. Phase difference due sources} = \pi$$

Fig. 8.4 below shows the displacement-time graph of the waves arriving at R when only transmitter A is switched on and when both transmitters A and B are switched on.

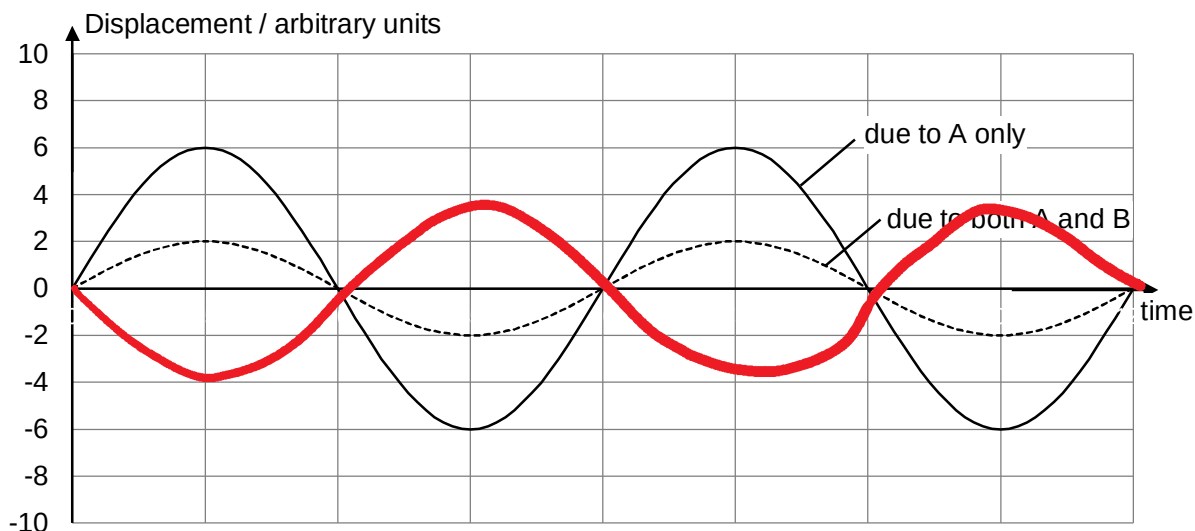


Fig. 8.4

- (iii) Sketch, on Fig. 8.4, the displacement-time graph of the waves arriving at R when only transmitter B is switched on. [1]

- (iv) Determine the ratio, at point R, $\frac{\text{intensity of microwave due to transmitter A}}{\text{intensity of microwave due to transmitter B}}$.

$$\frac{\text{Amplitude due to A}}{\text{Amplitude due to A and B}} = \frac{9}{3}$$

$$\frac{\text{Amplitude due to A}}{\text{Amplitude due to B}} = \frac{9}{6}$$

$$\frac{\text{Intensity due to A}}{\text{Intensity due to A and B}} = \left(\frac{9}{6}\right)^2 = 2.25$$

